

AI-Side Pluralsight Skill IQ Master Prep

One combined preparation guide for AI foundations, Agentic AI, Harness, ML evaluation, LLM evaluation, RAG evaluation, LLMOps, safety, and architecture.

Use this as a roadmap after completing the separate question banks. It is not a dump of official questions. It is designed to prepare you for adaptive, concept-heavy, scenario-based Skill IQ tests.

What this guide covers

Area	Why it matters for Skill IQ	Target level
AI and GenAI foundations	Tests often check if you can distinguish AI, ML, deep learning, generative AI, embeddings, tokens, model limitations, and responsible AI.	Medium
ML basics to advanced	Good scores require knowing models, feature engineering, overfitting, validation, and when metrics mislead.	Medium to hard
ML evaluation	You must know confusion matrix trade-offs, regression metrics, calibration, leakage, cross-validation, drift, and production monitoring.	Hard
LLM architecture	High-scorers need decoder-only vs encoder-only vs encoder-decoder, attention, causal masks, KV cache, context windows, LoRA, quantization, MoE.	Hard to very hard
Prompt engineering	Skill IQ-style questions often test prompt patterns, constraints, examples, temperature, structured output, injection, and evaluation.	Medium to hard
LLM and RAG evaluation	Modern AI tests increasingly ask about groundedness, faithfulness, retrieval recall, LLM-as-judge, regression tests, observability, and drift.	Hard
Agentic AI	Know tools, planning, memory, orchestration, multi-agent systems, permissioning, human-in-the-loop, failure modes, and safety.	Hard
Harness	Prepare for Harness.io path concepts and the broader idea of an agent harness: IDP, SEI, CDEs, DevOps automation, knowledge graphs, governance.	Medium to hard

Recommended scoring strategy

- Do not only memorize definitions. Skill IQ is adaptive, so once you answer basics correctly, questions become more scenario-based.
- Prioritize contrast questions: RAG vs fine-tuning, encoder-only vs decoder-only, accuracy vs F1, agent vs workflow, prompt engineering vs context engineering vs harness engineering.
- For high scores, answer from the perspective of production trade-offs: cost, latency, risk, observability, governance, and failure recovery.

1. Official scope signals to use

Pluralsight states that Skill IQ assessments are adaptive, short, and typically around 20-25 multiple-choice questions. The current AI-related paths cover AI Foundations, Generative AI Foundations, Prompt Engineering, Agentic AI for Developers, Integrating Agentic AI, Harness, and LLMOps topics. Use these paths as topic boundaries, not as exact question sources.

Source signal	What to prepare from it
Skill IQ format	Timed MCQs, adaptive difficulty, short but broad. Expect mixed definition, scenario, and trade-off questions.
AI Foundations	AI vs ML vs DL vs GenAI, model training, inputs/outputs, limitations, ethics, business use cases.
Prompt Engineering	Instructions, context, examples, constraints, structured output, model parameters, prompt injection.
Agentic AI paths	Agents that plan, act, iterate, coordinate, use tools, and operate with safety controls.
Harness path	Internal Developer Portal, Software Engineering Insights, Cloud Development Environments, productivity, quality, and software delivery optimization.
LLMOps evaluation course	Evaluation metrics, datasets, observability, logging, regression tests, drift detection, dashboards, canary testing.

Source notes appear at the end of this PDF.

2. AI foundations - what else to prepare

Topic	You should know	Common trap
AI vs ML vs DL vs GenAI	AI is the broader field; ML learns from data; deep learning uses neural networks; GenAI creates new content.	Saying all AI is generative AI.
Discriminative vs generative	Discriminative models separate/classify; generative models learn to produce samples or likely outputs.	Thinking generative always means LLM only.
Training data	Quality, bias, representativeness, labels, leakage, and governance affect performance.	Assuming more data always helps.
Tokens and embeddings	Tokens are model input units; embeddings are vector representations useful for semantic similarity.	Thinking a token is always a word.
Inference	Using a trained model to generate predictions or outputs.	Confusing inference with training.
Modalities	Text, image, audio, video, tabular, and multimodal inputs/outputs.	Assuming multimodal means only image input.
Responsible AI	Fairness, privacy, explainability, safety, accountability, transparency, data governance.	Treating ethics as optional instead of a system requirement.

High-yield distinctions

- Prediction vs generation: prediction outputs a label/value; generation produces content.
- Automation vs autonomy: automation follows predefined steps; autonomy can plan/select actions within boundaries.
- Model capability vs product capability: a product combines model, tools, data, UX, guardrails, monitoring, and permissions.
- Model knowledge vs retrieved knowledge: model weights store learned statistical patterns; RAG provides external context at runtime.

Mini-question patterns

- Which approach is best when sensitive data cannot leave the organization?
- Which risk is caused by non-representative training data?
- Which term describes converting text into dense vectors for semantic search?

3. ML basics to advanced - beyond the obvious

Area	Prepare	High-score clue
Supervised learning	Classification vs regression, labels, loss functions, train/validation/test split.	Know when classification metrics beat accuracy.
Unsupervised learning	Clustering, dimensionality reduction, anomaly detection, PCA, k-means, DBSCAN basics.	Know that cluster labels are not ground truth by default.
Feature engineering	Scaling, encoding, missing values, leakage, interaction features, text/vector features.	Data preprocessing must be fitted on train only.
Bias-variance	Underfitting vs overfitting, regularization, model complexity, validation curves.	High train score + low test score = overfit.
Tree models	Decision trees, random forests, gradient boosting, feature importance.	Trees do not require scaling in the same way as distance-based models.
Neural networks	Activation, loss, backprop, learning rate, batch size, epochs, early stopping.	More epochs can worsen generalization.
Model selection	Cross-validation, hyperparameter tuning, baselines, ablation, reproducibility.	A simple baseline is essential.

ML evaluation topics to master

Metric / concept	Use when	Trap
Accuracy	Balanced classes and equal error cost.	Misleading with class imbalance.
Precision	False positives are expensive.	High precision can miss many positives.
Recall	False negatives are expensive.	High recall can create many false positives.
F1	Need balance between precision and recall.	Hides different precision/recall trade-offs.
ROC-AUC	Ranking quality across thresholds.	Can look good on imbalanced data.
PR-AUC	Imbalanced positive class problems.	More informative than ROC-AUC for rare positives.
MAE	Regression with interpretable absolute errors.	Less sensitive to large errors than RMSE.
RMSE	Regression where large errors should be penalized strongly.	Outliers dominate.
Calibration	Predicted probabilities must mean real probabilities.	High accuracy does not imply calibrated probabilities.
Drift	Data or behavior changes after deployment.	Offline test performance can decay.

4. LLM architecture - high-value topics

Topic	Know this	Likely question angle
Decoder-only Transformers	GPT/Llama/Mistral-style models generate next tokens autoregressively using causal self-attention.	Why they can answer and translate despite no separate encoder.
Encoder-only Transformers	BERT-style models produce bidirectional representations for understanding tasks.	Best for classification/search, not open-ended generation.
Encoder-decoder Transformers	T5/BART/original Transformer style: encoder reads input, decoder generates output.	Translation/summarization architectures.
Causal mask	Prevents future tokens from being attended to during generation/training.	Why next-token prediction works.
Self-attention	Lets tokens weigh other relevant tokens in context.	Long-range dependency handling.
Multi-head attention	Learns multiple attention patterns in parallel.	Different heads can focus on syntax, entities, positions, etc.
Context window	Maximum tokens available in the current input/output context.	Long context does not guarantee perfect recall.
KV cache	Stores previous key/value states to speed up autoregressive decoding.	Improves inference latency but uses memory.
RoPE / positional encodings	Provide position information because attention alone is order-agnostic.	Why token order matters.
MoE	Mixture-of-experts activates a subset of expert networks per token.	Can increase capacity without activating all parameters.
Quantization	Reduces precision/size for cheaper inference.	Can reduce memory/cost but may affect quality.
LoRA / PEFT	Fine-tunes small adapter weights instead of all model weights.	Efficient customization.

Architectural mental model

A decoder-only LLM can still interpret a question because the prompt tokens are part of the same sequence. Attention creates internal representations of the prompt, and the model continues the sequence with the answer. Translation, summarization, classification, extraction, and reasoning can all be cast as next-token prediction when the prompt gives the task pattern.

5. Prompt engineering - beyond basic prompts

Topic	Prepare	High-score trap
Instruction hierarchy	System/developer/user/tool messages have different authority levels.	Treating user prompt as always highest priority.
Few-shot prompting	Examples establish pattern, format, and edge cases.	Bad examples can anchor wrong behavior.
Structured output	Use schemas, JSON constraints, validation, retries.	Prompt alone does not guarantee valid output.
Temperature/top-p	Sampling controls randomness/diversity.	Low temperature improves consistency, not truth.
Context engineering	Supplying the right context, metadata, retrieval, tools, and state.	Thinking prompts alone solve knowledge gaps.
Prompt injection	Untrusted content tries to override system instructions or leak data.	Treating retrieved documents as trusted instructions.
Tool prompting	Clear tool descriptions, argument schemas, permissions, and confirmation rules.	Letting model call dangerous tools without checks.
Evaluation	Test prompts using representative cases and failure cases.	Judging prompt quality from a single demo.

Prompt patterns worth knowing

- Zero-shot: direct instruction with no examples.
- Few-shot: include examples and expected outputs.
- Chain-of-thought style: ask for stepwise reasoning when appropriate, but do not expose private reasoning in production apps.
- Decomposition: split complex tasks into subtasks.
- Self-check / critique: ask the model to inspect outputs, but verify with external rules when correctness matters.
- RAG-grounded prompting: instruct the model to answer only from retrieved context and admit insufficient evidence.

6. RAG evaluation and LLM evaluation

Evaluation layer	Metrics / checks	What good looks like
Retriever	Recall@k, precision@k, MRR, nDCG, context relevance, duplicate rate.	Relevant evidence appears in the retrieved context before generation.
Chunking	Chunk size, overlap, metadata quality, boundary quality.	Chunks are self-contained enough for answering, but not too large/noisy.
Reranking	Reranker improves top results after vector search.	Useful when initial retrieval has recall but poor ordering.
Generator	Faithfulness, groundedness, answer relevance, completeness, citation correctness.	Answer is supported by context and does not invent unsupported facts.
System	Latency, cost, token usage, refusal quality, fallback behavior.	Meets quality targets within production constraints.
Safety	Prompt injection resistance, PII leakage, policy compliance.	Untrusted text is treated as data, not instruction.

LLM evaluation categories

Category	Examples	Notes
Task quality	Accuracy, relevance, completeness, style, format compliance.	Needs task-specific benchmark examples.
Truthfulness	Factuality, groundedness, citation correctness.	Often needs source checking or retrieval context.
Safety	Toxicity, bias, jailbreak resistance, privacy leakage.	Test both normal and adversarial prompts.
Robustness	Paraphrases, noisy input, edge cases, multilingual cases.	Small wording changes should not break behavior.
Operational	Latency, cost, throughput, cache hit rate, failure rate.	Essential for production but often ignored in demos.
Regression	Golden sets, release gates, prompt/model comparison.	Prevents silent degradation after changes.

7. Agentic AI - what to prepare

Topic	Prepare	Likely trap
Agent vs chatbot	An agent can choose actions/tools and progress toward goals; a chatbot mainly responds.	Calling every LLM app an agent.
Agent vs workflow	Workflow is predefined; agentic system has dynamic planning or tool selection.	Overusing agents when deterministic workflow is safer.
Tool/function calling	Schemas, argument validation, tool permissions, retries, error handling.	Assuming model-generated tool arguments are always valid.
Planning	ReAct, plan-and-execute, task decomposition, reflection, replanning.	Plans can be wrong; execution needs verification.
Memory	Short-term context, long-term/persistent memory, episodic/user memory, vector memory.	Memory can become stale or unsafe.
Orchestration	Single-agent vs multi-agent, routing, supervisors, specialists, DAGs, event-driven agents.	More agents often means more coordination failure.
Human-in-the-loop	Required for high-risk, irreversible, financial, legal, security, or external actions.	Fully autonomous action without approvals.
Safety	Sandboxing, permission boundaries, audit logs, rate limits, secret protection, confirmation prompts.	Giving the agent broad credentials.
Evaluation	Task completion, tool accuracy, cost, latency, safety violations, recovery from tool failure.	Only evaluating final text quality.

Agentic concepts to memorize

- **Autonomy:** ability to make decisions within boundaries.
- **Tool use:** calling APIs, databases, files, browsers, calendars, code execution, or external systems.
- **Observation-action loop:** observe state, decide action, call tool, observe result, continue or stop.
- **Delegation:** routing a subtask to a specialist agent or tool.
- **Guardrail:** control that constrains or validates model behavior before/after actions.
- **Agent harness:** runtime layer around a model that provides tools, memory, policies, monitoring, evaluation, and orchestration.

8. Harness - prepare for both meanings

Harness can mean the Harness.io software delivery platform, or more generally an agent harness/runtime that wraps an LLM with tools, memory, policies, monitoring, and verification. For Pluralsight, the Harness path is about Harness.io concepts; for AI discussions, agent harness is a broader architecture concept.

Harness area	Prepare	Why it matters
Internal Developer Portal (IDP)	Catalogs, scorecards, templates, onboarding, service ownership, golden paths.	Improves developer self-service and standardization.
Software Engineering Insights (SEI)	DORA metrics, flow metrics, dashboards, team/collection views, productivity and quality measurement.	Turns engineering data into decisions.
Cloud Development Environments (CDEs)	Secure, consistent, browser/cloud-based dev environments, faster onboarding, reduced local setup drift.	Standardizes dev environments.
Harness AI	DevOps automation, testing, security, FinOps, SRE, release agents, pipeline troubleshooting.	Focus is after-code software delivery, not just code generation.
Knowledge graph	Connects builds, tests, deployments, incidents, infrastructure, cloud spend, and delivery context.	Provides context for AI recommendations.
Governance	RBAC, guardrails, audit trails, approvals, policy enforcement.	Essential when AI can affect delivery systems.
DORA metrics	Deployment frequency, lead time for changes, change failure rate, failed deployment recovery time / MTTR.	Common metric questions in DevOps/SEI contexts.

Harness-style scenario answers

- If the problem is lack of service ownership or poor onboarding: think IDP/catalog/templates/scorecards.
- If the problem is engineering visibility or productivity metrics: think SEI/DORA/flow metrics/dashboards.
- If the problem is inconsistent local environments: think CDEs.
- If the problem is pipeline troubleshooting or release automation: think Harness AI/DevOps Agent/workflow orchestration.
- If the problem is safe enterprise adoption: think RBAC, audit logs, guardrails, policy-as-code, approval gates.

9. LLMOps and AI production readiness

Area	Prepare	Skill IQ angle
Model selection	Accuracy, cost, latency, context, modality, privacy, licensing, vendor lock-in.	Best model is not always largest model.
Deployment	API vs self-hosted, batching, autoscaling, concurrency, rate limits.	Trade-offs, not absolutes.
Caching	Prompt/result cache, embedding cache, retrieval cache.	Reduces cost/latency but can serve stale outputs.
Observability	Logs, traces, prompts, outputs, tool calls, token usage, user feedback.	Need privacy-aware logging schemas.
Evaluation pipeline	Offline evals, online monitoring, canaries, A/B tests, regression sets.	Eval should run continuously, not once.
Drift	Input drift, behavior drift, retrieval corpus drift, model version drift.	GenAI systems drift even when model unchanged.
Security	Secrets, prompt injection, data exfiltration, dependency/tool risk.	Tools expand attack surface.
Governance	Data retention, access control, auditability, responsible AI policies.	Enterprise AI needs controls, not only accuracy.

Best-practice production principle

For Skill IQ scenarios, the safest answer is often not 'use a bigger model'. It is usually: improve data quality, retrieval quality, evaluation coverage, monitoring, guardrails, and human review for high-risk actions.

10. High-score test traps

Trap	Correct response
High accuracy on imbalanced data	Check precision, recall, F1, PR-AUC, confusion matrix.
RAG answers unsupported by context	Improve retrieval, grounding instructions, citation validation, and faithfulness checks.
User asks model to ignore system rules	Identify prompt injection; follow instruction hierarchy.
Frequent hallucinations in enterprise assistant	Add/strengthen RAG, restrict answer scope, evaluate groundedness, improve source quality.
Need model to know changing policies	Prefer RAG or tool retrieval over fine-tuning.
Need specific style/format behavior	Prompting or fine-tuning may help; still validate output.
Need lower inference cost	Consider smaller model, quantization, caching, batching, routing, or distillation.
Agent has too many tool failures	Add tool schemas, validation, retries, error recovery, tool-specific tests.
Multi-agent system performs worse	Simplify orchestration; reduce coordination overhead; use deterministic workflow where possible.
ML model worked offline but failed in production	Check leakage, drift, data pipeline mismatch, monitoring, and retraining strategy.
Harness vs agent harness confusion	Harness.io is a software delivery platform; agent harness is a broader LLM runtime pattern.

11. Priority ranking for your preparation

Priority	Topic	Why
1	LLM architecture and RAG	You are already strong in RAG, so make this your scoring advantage. Add architecture precision.
2	LLM evaluation and ML evaluation	Evaluation questions separate average learners from high scorers.
3	Agentic AI design and safety	Newer Skill IQ areas increasingly test tools, orchestration, memory, and governance.
4	Harness.io fundamentals	The Harness Skill IQ path is short, so small focused prep can yield quick gains.
5	Prompt engineering and injection	Common across many AI-related Skill IQs.
6	AI foundations and ethics	Usually easier marks; do not skip responsible AI.
7	MLOps/LLMOps production readiness	Important for scenario questions about deployment, monitoring, and trade-offs.

14-day plan

Days	Focus	Output
1-2	AI foundations, GenAI basics, tokens, embeddings, responsible AI.	One-page terminology sheet.
3-4	ML metrics, validation, leakage, overfitting, drift.	Confusion matrix and metric decision table.
5-6	LLM architecture: Transformers, attention, causal masks, KV cache, model types.	Architecture comparison table.
7-8	Prompt engineering, structured output, injection, RAG prompting.	Prompt pattern examples and failure cases.
9-10	RAG and LLM evaluation: recall@k, faithfulness, LLM-as-judge, regression tests.	Evaluation checklist for your RAG project.
11-12	Agentic AI: tools, memory, orchestration, guardrails, human-in-loop.	Agent design checklist.
13	Harness: IDP, SEI, CDEs, DORA, Harness AI, governance.	Harness scenario map.
14	Timed mixed practice.	Retake weak sections only.

12. Mixed advanced practice questions

Use these as Skill IQ-style practice. They are intentionally scenario-heavy and not official Pluralsight questions.

Q1. A fraud model has 99% accuracy but misses most fraud cases. Which metric should you inspect first?

A. Training loss only B. Recall and PR-AUC C. Mean squared error D. BLEU

Answer: B. Fraud is usually imbalanced and false negatives matter; recall and PR-AUC are more informative than raw accuracy.

Q2. A RAG chatbot often gives answers unrelated to the retrieved documents. Which evaluation category catches this?

A. Faithfulness / groundedness B. Temperature C. Parameter count D. Tokenization speed

Answer: A. Groundedness checks whether the answer is supported by context.

Q3. A GPT-style model can translate text even without a separate encoder because...

A. it secretly uses CNNs B. translation can be represented as next-token completion conditioned on the prompt C. embeddings are the final output D. causal masks are removed at inference

Answer: B. Decoder-only models condition on prompt tokens and generate the continuation.

Q4. You need current company policy answers with citations. Best first approach?

A. Fine-tune on old policies B. Use RAG with current policy documents C. Raise temperature D. Use only a larger model

Answer: B. Changing knowledge and evidence requirements are classic RAG use cases.

Q5. A model's probabilities are overconfident despite good classification accuracy. What should you assess?

A. Calibration B. Token count C. MoE routing D. Embedding dimension only

Answer: A. Calibration measures whether predicted probabilities match real likelihood.

Q6. Which best describes an agent harness?

A. A dataset format B. Runtime around a model with tools, memory, policies, monitoring, and orchestration C. Only a CI/CD vendor D. A type of loss function

Answer: B. The technical term refers to the infrastructure that makes an LLM usable as an agent.

Q7. For an enterprise tool-calling agent, the model returns malformed JSON arguments. Best mitigation?

A. Increase temperature B. Add schema validation and retry/repair logic C. Remove logs D. Disable all tools forever

Answer: B. Tool arguments should be validated before execution.

Q8. Which is most likely to improve retrieval ordering after vector search already finds relevant candidates?

A. Reranking B. Lowering batch size C. Increasing epochs D. Using MAE

Answer: A. Rerankers reorder candidate chunks by deeper relevance.

Q9. What does a causal mask do in decoder-only training?

A. Hides previous tokens B. Prevents attending to future tokens C. Encrypts model weights D. Converts text to vectors

Answer: B. Causal masks preserve next-token prediction behavior.

Q10. A team wants to reduce LLM inference memory with possible small quality loss. Which technique fits?

A. Quantization B. PCA on labels C. SMOTE D. ROC curve

Answer: A. Quantization reduces precision and memory footprint.

Q11. Which Harness concept best fits measuring DORA and flow metrics across teams?

A. Software Engineering Insights B. Cloud Development Environment C. Prompt temperature D. Tokenizer

Answer: A. SEI is focused on engineering metrics and dashboards.

Q12. A model has high training accuracy and low validation accuracy. Main issue?

A. Underfitting B. Overfitting C. Perfect calibration D. Low context window

Answer: B. The model memorized training data but generalizes poorly.

Q13. Which metric is most directly about retrieval quality, not answer generation?

A. Recall@k B. Toxicity C. Latency percentile D. Rouge-L

Answer: A. Recall@k checks whether relevant items are in the top k retrieved results.

Q14. An LLM system regresses after changing the prompt template. What should have caught this?

A. Golden-set regression evaluation B. More GPU memory only C. Training accuracy D. A larger chunk size always

Answer: A. Regression sets compare outputs across versions.

Q15. A user asks the model to reveal hidden instructions from a retrieved document. The document says 'ignore all previous rules'. What is this?

A. Normal context B. Prompt injection C. Calibration D. Clustering

Answer: B. Untrusted retrieved text is trying to override instructions.

Q16. Which scenario most clearly needs human-in-the-loop approval?

A. Summarizing public docs B. Deleting production resources C. Rephrasing a sentence D. Counting tokens

Answer: B. Irreversible/high-risk external actions need approval.

Q17. What is a KV cache used for?

A. Faster autoregressive decoding B. Label smoothing C. Data balancing D. Feature normalization

Answer: A. It stores key/value states from previous tokens.

Q18. Which model type is best associated with BERT-style representation learning?

A. Encoder-only B. Decoder-only C. CNN-only D. Decision tree

Answer: A. BERT is encoder-only and bidirectional.

Q19. A model output follows the wrong JSON format. Which eval dimension catches this?

A. Format compliance B. ROC-AUC C. K-means inertia D. Lead time

Answer: A. Structured output tasks require schema/format compliance.

Q20. For a rare disease screening model, missing true positives is very costly. Which metric should be prioritized?

A. Recall / sensitivity B. Specificity only C. RMSE D. BLEU

Answer: A. Recall measures how many actual positives are detected.

Q21. Which is the best reason to use a smaller model in production?

A. It is always more accurate B. It may meet quality requirements with lower cost/latency C. It avoids all hallucinations D. It removes need for monitoring

Answer: B. Model selection is a trade-off.

Q22. Which Agentic AI failure mode comes from giving an agent broad credentials and no approval gates?

A. Excessive autonomy risk B. Higher PR-AUC C. Data normalization D. Better calibration

Answer: A. Over-permissioned agents can take unsafe actions.

Q23. A vector search retrieves semantically similar but outdated policy chunks. What should you improve?

A. Metadata filters and freshness handling B. Temperature only C. CNN kernel size D. Accuracy

Answer: A. RAG retrieval should account for metadata like dates and versions.

Q24. What is the best description of MoE?

A. Every parameter active for every token B. A subset of expert networks is activated per token C. A confusion matrix metric D. A prompt injection method

Answer: B. Mixture-of-experts routes tokens to selected experts.

Q25. Which Harness concept best addresses inconsistent local developer setup?

A. CDEs B. BLEU C. Top-p D. Calibration

Answer: A. Cloud Development Environments standardize dev setup.

Q26. What does LLM-as-judge evaluate?

A. It uses another/model prompt to assess outputs against criteria B. It only counts tokens C. It trains all weights D. It replaces all human review

Answer: A. LLM judges can grade outputs, but should be validated and calibrated.

Q27. Which is the best first fix for noisy retrieval context causing hallucinations?

A. Improve chunking, metadata, retrieval, and reranking B. Increase temperature C. Remove sources D. Use accuracy metric only

Answer: A. Poor context often causes weak grounded answers.

Q28. Which is an example of data leakage?

A. Fitting scalars only on train data B. Using future information in training features C. Evaluating on held-out data D. Logging predictions

Answer: B. Leakage happens when unavailable/future/test information influences training.

Q29. What does an Internal Developer Portal mainly help with?

A. Developer self-service, service catalog, ownership, templates B. Image segmentation C. Token sampling D. LLM perplexity only

Answer: A. IDPs improve platform engineering and developer experience.

Q30. A classifier threshold is changed. Which metrics can shift even if model ranking is unchanged?

A. Precision and recall B. Tokenizer vocabulary C. Context window D. Number of parameters

Answer: A. Threshold changes affect predicted classes and confusion matrix metrics.

Q31. Why is long context not a complete replacement for RAG?

A. Long context guarantees perfect use of all info B. It can be costly/noisy and still lacks freshness/filtering/citation pipeline C. It prevents prompt injection D. It removes evaluation needs

Answer: B. Long context helps but does not solve retrieval, freshness, governance, or eval.

Q32. Which is best for testing prompt robustness?

A. Only one happy-path example B. Paraphrases, edge cases, adversarial inputs, and representative examples C. No examples D. Only token counts

Answer: B. Robustness requires varied and adversarial tests.

Q33. What is the main purpose of early stopping?

A. Stop training when validation performance stops improving B. Increase parameter count C. Disable validation D. Convert model to a vector DB

Answer: A. Early stopping is a regularization/generalization strategy.

Q34. Which answer best compares agent and deterministic workflow?

A. Agents are always safer B. Workflows are predefined; agents can dynamically plan/select actions C. Workflows require LLMs D. Agents cannot use tools

Answer: B. The distinction is autonomy and dynamic action selection.

Q35. In RAG, what does MRR measure?

A. Rank position of first relevant result B. Toxicity score C. Cloud cost D. Calibration slope

Answer: A. Mean Reciprocal Rank rewards relevant results appearing earlier.

Q36. What is a canary rollout useful for in LLMOps?

A. Gradually exposing a new model/prompt to limited traffic B. Training a CNN C. Removing evaluation D. Increasing future-token attention

Answer: A. Canaries reduce deployment risk.

Q37. Which is the strongest eval design?

A. Only synthetic easy examples B. Representative real cases plus edge/adversarial cases and clear rubrics C. No human review D. Judge by vibes

Answer: B. Evaluation datasets must represent real and risky cases.

Q38. If false positives are more expensive than false negatives, prioritize...

A. Precision B. Recall C. RMSE D. Context length

Answer: A. Precision focuses on correctness among predicted positives.

Q39. Which best describes top-p sampling?

A. Selects from smallest set of tokens whose cumulative probability reaches p B. Trains a model C. Splits documents D. Measures F1

Answer: A. Nucleus sampling controls the candidate probability mass.

Q40. What should you log for LLM observability while respecting privacy?

A. Nothing ever B. Prompts/outputs/tool calls/latency/tokens with redaction and retention controls C. All secrets D. Only model name

Answer: B. Useful observability must be privacy-aware.

13. Final checklist before taking Skill IQs

- Explain AI vs ML vs DL vs GenAI without mixing them.
- Choose correct ML metrics for imbalanced, high-risk, or regression problems.
- Recognize overfitting, leakage, drift, calibration, and threshold trade-offs.
- Explain encoder-only, decoder-only, encoder-decoder, attention, causal masks, context windows, KV cache, quantization, MoE, and LoRA.
- Decide when to use RAG, fine-tuning, prompt engineering, or tools.
- Evaluate RAG with retrieval metrics and answer faithfulness metrics.
- Design safe tool-calling agents with permissions, validation, audit logs, and human approval.
- Explain agent vs chatbot vs deterministic workflow vs multi-agent orchestration.
- Understand Harness IDP, SEI, CDEs, Harness AI, DORA metrics, knowledge graph, and governance.
- Answer scenario questions by balancing quality, cost, latency, privacy, safety, and maintainability.

Last-minute revision order

If you have only a few hours: review (1) ML metrics and leakage, (2) LLM architecture comparisons, (3) RAG/LLM evaluation, (4) agent tool safety and orchestration, (5) Harness IDP/SEI/CDE/DORA, and (6) prompt injection/guardrails.

Sources and scope notes

These sources were used to align the guide with current public Pluralsight and Harness topic scope. The practice questions in this PDF are original and are not official Pluralsight questions.

Topic	Source
Skill IQ format	Pluralsight Help Center: Introduction to Skill IQ; Technical skills assessments pages. Skill IQ assessments are adaptive, timed, and around 20-25 multiple-choice questions.
AI Foundations	Pluralsight AI: Foundations path. Covers core AI concepts and practical fundamentals.
Prompt Engineering	Pluralsight Prompt Engineering path. Covers crafting inputs for precise, context-aware outputs.
Agentic AI	Pluralsight Agentic AI in Industry; Agentic AI for Developers; Integrating Agentic AI for Developers paths.
Foundations of Agentic AI	Pluralsight professional services page covering chatbots, autonomous problem-solving agents, Hugging Face, LlamaIndex, and OpenAI Gym.
Harness	Pluralsight Harness path covering Internal Developer Portal, Software Engineering Insights, and Cloud Development Environments.
Harness AI	Harness AI official product page covering DevOps automation, testing/resilience, security/compliance, FinOps, knowledge graph, agentic flows, and governance.
LLMOps and evaluation	Pluralsight LLMOps: Evaluation, Observability, and Quality course outline covering evaluation metrics, datasets, observability, regression testing, drift detection, dashboards, and canary testing.